

Graded Containment with an Observability Price: Sequential Response to Ambiguous Trading Incidents

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Abstract

When an anomaly appears on an electronic trading gateway, the operator often cannot tell whether it comes from a compromise or from a fault, and the usual response is a kill-switch: stop the flow once the evidence is strong enough. We model the response as a finite-horizon partially observed Markov decision process over the hidden cause, with a graded ladder of actions (monitor, throttle, verify on a separate channel, kill, leave) and a belief that is updated from order flow and from the verification outcome. Throttling reduces harm but also blinds: it multiplies the per-period order-flow separation d by a discriminability factor $\kappa \in [0, 1]$. The model is solved exactly by backward induction on a belief grid, with a two-state baseline (compromise, fault) and a three-state extension in which a benign market shock is a third candidate cause. We prove that the value function is concave in the belief, that the kill and leave regions are intervals, that the optimal value is monotone in κ and in d , so that there is a single critical discriminability factor κ^* below which an optimal operator need never throttle, and that the one-shot verification rule of our companion paper is recovered exactly when order flow is uninformative. In a linear-Gaussian model of the gateway, a throttle that suppresses the incident's feedback into the book from the onset gives $\kappa = 0$ for pairs of causes confounded at the output level, and a throttle that scales the injection gives κ equal to the scaling factor. A pre-registered exact computation, calibrated on the low-prior cells of the companion identifiability study, tests two further claims: that the graded policy has strictly lower expected cost

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than the best Bayesian kill-switch; and that containment has an observability price, with κ^* strictly inside $(0, 1)$ and an operator who plans as if throttling kept full discriminability doing worse than the kill-switch.

1 Introduction

An alarm on an electronic trading gateway rarely comes with its cause attached. The same burst of abnormal orders can be the work of an adversary acting through a legitimate session (a *compromise*) or the symptom of a benign malfunction (a *fault*). The two call for opposite responses: a compromise should be contained quickly, while containing a fault disrupts a legitimate client for nothing. The operator’s standard tool is a kill-switch, which stops the flow once the evidence crosses a threshold. Between doing nothing and killing, operators also have intermediate options: they can throttle the flow, which limits harm while the incident continues, or check the activity on a channel that order flow does not carry, for instance by validating the session’s credentials or calling the client.

This paper asks when such a graded response is worth more than the best kill-switch, and what it costs. The answer turns on a feature that a static decision rule cannot see.

Containment changes what the operator observes next. Paper 1 [3] gives the starting point. When the incident’s injection enters the latent state of the book, Paper 1’s Theorem 2 shows that two mechanism instances are observationally indistinguishable at every time after the onset if and only if their injection difference $\Delta_0 = \theta_K b_K - \theta_F b_F$ lies in the unobservable subspace of (A, C) . Its finite-horizon version gives the same statement over H post-onset observations, with the condition $CA^j \Delta_0 = 0$ for $j = 0, \dots, H - 1$. A pair that gives the same immediate output ($C\Delta_0 = 0$, condition $(*)$ of Paper 1’s Theorem 1) can therefore still be told apart through the feedback of the injection into the state. Theorem 2 is an equivalence about equality of laws. It contains no containment action and no measure of how much discriminability is kept. Both come from Paper 3. We model a throttle that suppresses feedback as one that keeps the injection in the observation and removes it from the state recursion: throttled periods follow Paper 1’s output-level model (Definition 11). Proposition 4 then shows three things:

- for a pair confounded at the output level, such a throttle in force from the onset gives both causes the same law of observations. All the pair’s discriminability came from feedback, so the discriminability factor is

$\kappa = 0$.

- if the throttle starts later, a residual term remains, and it decays with the dynamics.
- a throttle that scales the injection by r gives $\kappa = r$. For other pairs, the linear model does not determine whether κ is below or above one.

That throttling costs discriminability ($\kappa \leq 1$, Assumption 4) is therefore an assumption of Paper 3’s model. It is derived from Paper 1’s model for the two throttles of Definition 11 and is not a prediction of Paper 1.

We study the response as a finite-horizon partially observed Markov decision process (POMDP) over the hidden cause (Section 3). The operator holds a belief that the cause is a compromise and, at each period, chooses one of five actions: *monitor*, *throttle*, *verify* (the costed, delayed check of Paper 2 [4]), *kill* or *leave*. The model is solved exactly by backward induction on a belief grid, without reinforcement learning or function approximation. In the baseline the cause is a compromise or a fault; in the extension a benign market shock is a third candidate cause, and the belief lives on the 2-simplex.

Contributions.

- *Model*. A POMDP in which containment has two effects at once: it scales the harm rate by a harm factor ρ and the per-period order-flow separation by the discriminability factor κ (Definitions 1–10, Assumptions 1–5). Both are defined before use in Section 3.
- *Where $\kappa < 1$ comes from* (Proposition 4, Section 3.7): the two throttles of Definition 11 in Paper 1’s linear-Gaussian model.
- *Structure* (Section 4): concavity of the value function (Lemma 1, Proposition 1); monotonicity of the optimal value in κ and in d , which makes the critical discriminability factor κ^* a single threshold below which an optimal operator need never throttle (Theorem 1); kill and leave regions that are intervals (Proposition 2); and the exact embedding of Paper 2’s one-shot rule when order flow is uninformative (Proposition 3).
- *A pre-registered numerical test* (Section 5) of the paper’s two substantive claims, on a region calibrated from Paper 1’s low-prior cells: that the graded policy has strictly lower expected cost than the best Bayesian kill-switch, and that containment has an observability price.

The design, the tolerance, the materiality margin and the criteria that would weaken or refute each claim are fixed before any result exists.

The comparison is against the *best* Bayesian kill-switch, an optimal stopping rule on the belief, not a fixed threshold. Weak dominance of the graded policy holds by construction, since the kill-switch is in its policy class; it is not claimed as a finding. What the paper claims is *strict* dominance on a stated region, with the gain split into what verification and what throttling contribute. Section 2 positions the paper; Section 6 discusses scope and limits.

2 Related work

POMDP intrusion response. The closest precedent is the treatment of intrusion prevention as an optimal stopping problem by Hammar and Stadler [1]: a defender with several progressively stronger stops faces a partially observed intrusion, and the optimal policies are thresholds in the belief. Our setting differs in three ways. First, the hypotheses there are intrusion against no intrusion, whereas ours are compromise against fault (and, in the extension, a benign shock): both hypotheses are anomalies, and the question is their cause. Second, there is no verification on a separate channel. Third, a defensive action there does not degrade what the defender observes next, while in our model throttling multiplies the order-flow separation by κ . The same authors extend the stopping formulation to a dynamic attacker and learn near-optimal responses by self-play [2]; our cause is non-strategic (Assumption 5). Miehl, Rasouli and Teneketzis [5] model the defence of a network against an attacker progressing on an attack graph as a POMDP, in which the defender trades security against availability. There the belief is over the attacker’s progress; here it is over the cause of an anomaly, a scalar or a point of the 2-simplex, and the operator’s actions change the quality of observation. Automated response engines that pair attack trees with a partially observable decision process [6] likewise take the attack as given and ask how to respond to it, not whether an attack is present.

Verification and one-shot diagnosis. The verify action is the costed, delayed check of Paper 2 [4], which chooses once between leaving, verifying and containing. Our model makes that choice sequential and adds order flow and throttling; Proposition 3 shows that Paper 2’s rule is recovered exactly when order flow is uninformative and the horizon equals the verification delay.

Active hypothesis testing and dual control. Choosing observations to learn a hypothesis is the subject of the sequential design of experiments and of active and controlled-sensing hypothesis testing. There, actions buy information. In our model the most informative posture, **monitor**, is also the most harmful under a compromise, and the action that reduces harm, **throttle**, reduces information: the throttle has a dual role through ρ and κ together. That the value of an experiment is monotone in its informativeness is Blackwell’s ordering; Theorem 1 is an instance of this technique applied to the throttle’s discriminability factor, and we do not claim the technique as new. What the paper adds is where the resulting threshold κ^* lies on a calibrated region and what an operator loses by ignoring it.

Kill-switches and circuit breakers. Kill-switch and circuit-breaker practice on trading venues stops activity on a fixed trigger. Our kill-switch baselines (Definition 9) are the fixed-threshold rule KS_θ and the best Bayesian stopping rule KS^* ; the latter is the strongest version of the kill-switch that the model allows, and it is the one the graded policy is compared with.

3 Model

An operator watches an anomaly on a trading gateway. The anomaly has one of two causes, a compromise K or a fault F ; in the extension, a third cause B , a benign market shock, is also possible. The operator does not see the cause. At each period the operator observes an order-flow statistic and chooses an action, until a terminal action ends the episode or the horizon is reached. This section specifies the decision problem completely: time and states, actions, observations, verification, belief update, costs, the Bellman recursion and its numerical solution, the baseline policies, and the quantities the paper reports.

3.1 Time, causes and actions

Definition 1 (Time, hidden cause, model variants). Time is discrete, $t = 0, 1, \dots, T$, with horizon $T \geq 1$; decisions are taken at $t = 0, \dots, T - 1$ and the incident is under way from $t = 0$ (the alarm time). The hidden cause M is drawn once at $t = 0$ and does not change. In the *two-state model* $M \in \mathcal{M}_2 = \{K, F\}$ (compromise, fault) and the belief is the scalar $p_t = P(M = K \mid \mathcal{I}_t) \in [0, 1]$, where $\mathcal{I}_t$ is the history of actions and observations before the decision at t . In the *three-state model* $M \in \mathcal{M}_3 = \{K, F, B\}$,

where B is a benign market shock (no compromise, no fault), and the belief is the point $\pi_t = (\pi_t(K), \pi_t(F), \pi_t(B))$ of the 2-simplex Δ_2 . The prior is p_0 (resp. π_0).

Definition 2 (Actions). At each decision time the operator chooses one action from

$$\mathcal{A} = \{\text{monitor, throttle, verify, kill, leave}\}.$$

monitor and **throttle** are *postures* held for one period; **verify** is the verification action of Paper 2 (Definition 2 there), available at most once per episode, which holds the **monitor** posture for $\ell_v \geq 1$ periods and returns its outcome at the end of them; **kill** (contain) and **leave** (stand down) are *terminal* and end the episode. At $t = T$ only **kill** and **leave** are available. A *verification flag* $v_t \in \{0, 1\}$ records whether verification has been used; the decision state is (t, p_t, v_t) (resp. (t, π_t, v_t)).

The ladder has two postures and two terminal actions. **monitor** lets trading continue while the operator watches; **throttle** limits the flow (for instance by rate limits or reduced order sizes) at a disruption cost, reducing harm; **verify** queries a channel that order flow does not carry; **kill** contains (suspends the session, pulls the orders); **leave** stands down. Leave is needed both so that the kill-switch baselines can stand down as well as kill, and for the exact embedding of Paper 2’s rule (Proposition 3).

3.2 Observations and the discriminability factor

Definition 3 (Order-flow observations and separation). In each period spent in a posture $x \in \{\text{monitor, throttle}\}$ the operator observes an order-flow statistic $y \in \mathbb{R}$. In the two-state model,

$$y \mid (M = K, x) \sim \mathcal{N}(+\tfrac{1}{2}d_x, 1), \quad y \mid (M = F, x) \sim \mathcal{N}(-\tfrac{1}{2}d_x, 1), \quad d_{\text{monitor}} = d, \quad d_{\text{throttle}} = \kappa d,$$

with per-period separation $d \geq 0$ and *discriminability factor* $\kappa \geq 0$. In the three-state model, $y \in \mathbb{R}^k$ ($k \in \{1, 2\}$) with $y \mid (M, x) \sim \mathcal{N}(g_x \mu_M, I_k)$, $g_{\text{monitor}} = 1$, $g_{\text{throttle}} = \kappa$, for fixed means $\mu_K, \mu_F, \mu_B \in \mathbb{R}^k$; the pairwise per-period separations are $d_{MM'} = \|\mu_M - \mu_{M'}\|$ under monitoring and $\kappa d_{MM'}$ under throttling. The two-state model is the case $\mu_K - \mu_F = d$ of the three-state model restricted to $\{K, F\}$.

The factor κ is the share of the per-period separation that throttling keeps: in Paper 1’s linear model it is the ratio $\kappa_W = d_W(\Theta)/d_W(\emptyset)$ of whitened separations with and without the throttle (Definition 11).

The separation d is the distance between the two means in units of the noise. Throttling moves the two means towards each other by the factor

κ . At $\kappa = 1$ throttling costs no evidence, and at $\kappa = 0$ a throttled period carries no evidence at all.

Definition 4 (Verification outcome). The outcome $Z \in \{+, -\}$ of verify has the law of Paper 2, Definition 2: $P(Z = + | K) = s$, $P(Z = - | F) = t$. In the three-state model $P(Z = - | B) = t_B$, with $t_B = t$ unless stated otherwise (verification reads authorization, which is valid under both F and B). During the ℓ_v periods of verification the operator also observes the order-flow statistics of the **monitor** posture if $\omega = 1$ and none if $\omega = 0$; $\omega = 0$ is the analogue of Paper 2, Assumption 4 (no further order-flow evidence during the delay).

Definition 5 (Belief update). After an order-flow statistic y observed in posture x , the two-state belief is updated by Bayes' rule,

$$\text{logit } p' = \text{logit } p + d_x y,$$

(the log-likelihood ratio of the two Gaussians of Definition 3), and after a verification outcome z by $p' = p^z$ with p^+, p^- as in Paper 2, Definition 2. In the three-state model, $\pi'(M) \propto \pi(M) \phi_k(y - g_x \mu_M)$ after y and $\pi'(M) \propto \pi(M) P(Z = z | M)$ after z , where ϕ_k is the standard normal density on $\mathbb{R}^k$.

3.3 Costs and optimal value

Definition 6 (Costs). Costs use the general loss table of decision 0014, extended to B in the three-state model: **kill** at any t costs C_M and ends the episode; **leave** at any t costs H_M and ends it. Each period spent in a posture adds a stage cost: **monitor** costs r_M (harm accrued in the period), **throttle** costs $\delta + \rho r_M$, with disruption cost $\delta \geq 0$ and *harm factor* $\rho \in [0, 1]$; **verify** costs c_v once plus r_M for each of its ℓ_v periods. The baseline harm rates are $r_K \geq 0$ and $r_F = r_B = 0$. Stage costs are additional to the terminal cost, whichever terminal action ends the episode, and there is no discounting.

The throttle trades a certain disruption cost δ against a reduction of expected harm from pr_K to ρpr_K , and it pays for this reduction with evidence, through κ . Whether that trade is worth making is what the value of throttling of Definition 10 measures.

Definition 7 (Policies, value, Bellman recursion). A policy maps each decision state to an action. Its *expected cost* $J^\pi(p_0)$ is the expected sum of stage and terminal costs from prior p_0 with $v_0 = 0$. In the two-state model the optimal value $V_t(p, v)$ satisfies $V_T(p, v) = R(p) = \min\{pC_K + (1 - p)C_F, pH_K + (1 - p)H_F\}$ and, for $t < T$,

$$V_t(p, v) = \min_{a \in \mathcal{A}_t(v)} Q_t(p, v, a),$$

$$\begin{aligned}
Q_t(p, v, \text{kill}) &= p C_K + (1 - p) C_F, & Q_t(p, v, \text{leave}) &= p H_K + (1 - p) H_F, \\
Q_t(p, v, \text{monitor}) &= p r_K + \mathbb{E}_y[V_{t+1}(p', v)], \\
Q_t(p, v, \text{throttle}) &= \delta + \rho p r_K + \mathbb{E}_y[V_{t+1}(p', v)], \\
Q_t(p, 0, \text{verify}) &= c_v + \ell_v p r_K + \mathbb{E}[V_{t+\ell_v}(p'', 1)],
\end{aligned}$$

where p' is the update of Definition 5 in the posture taken, the expectation is over y under the predictive mixture $p\mathcal{N}(\frac{1}{2}d_x, 1) + (1 - p)\mathcal{N}(-\frac{1}{2}d_x, 1)$, p'' is the belief after the verification outcome (and, if $\omega = 1$, the ℓ_v monitoring observations), $\mathcal{A}_t(1)$ excludes **verify**, and **verify** is excluded when $t + \ell_v > T$. The three-state recursion is the same with p replaced by $\pi \in \Delta_2$ and every $p X_K + (1 - p) X_F$ by $\sum_M \pi(M) X_M$. Ties are broken towards the lower rung of the order **leave**, **monitor**, **verify**, **throttle**, **kill**.

3.4 Numerical solution

Definition 8 (Belief grid). The recursion of Definition 7 is solved by backward induction on a grid: in the two-state model the uniform grid $G_N = \{i/N : i = 0, \dots, N\}$, with V_{t+1} evaluated off the grid by piecewise-linear interpolation and $\mathbb{E}_y$ computed by a Bayes-consistent quadrature rule of order n_q (nodes and weights independent of p , Proposition 1(iii)); in the three-state model the triangular grid $G_N^{(2)} = \{\pi \in \Delta_2 : N\pi \in \mathbb{Z}^3\}$ with barycentric linear interpolation and $\mathbb{E}_y$ computed by Gauss–Hermite quadrature of order n_q in each Gaussian component (the *per-component Gauss–Hermite rule*, also a stated variant in the two-state model). The reported value is the grid value; a solution is *grid-converged at tolerance ϵ* if doubling N and n_q changes $V_0(p_0, 0)$ by at most ϵ at every reported prior. No reinforcement learning or function approximation is used.

The two-state recursion is solved with a Bayes-consistent quadrature rule, under which the grid problem is itself an exact Bayes recursion for a discrete observation; concavity of the grid values (Proposition 1(iii)) is then a checkable property of the solver’s output, used as a test of the implementation. The three-state recursion uses the per-component Gauss–Hermite rule of Definition 8. The solver is written in NumPy; the code, with the version used for every reported computation, is to be made available in an archived repository.

3.5 Baselines and reported quantities

Definition 9 (Baseline policies). (i) *Fixed-threshold kill-switch* KS_θ : monitor while $p_t < \theta$, kill as soon as $p_t \geq \theta$, and at $t = T$ the better terminal

action at p_T . (ii) *Best Bayesian kill-switch* KS^* : the optimal policy of the recursion of Definition 7 with the action set restricted to $\{\text{monitor}, \text{kill}, \text{leave}\}$ (an optimal stopping rule on the belief, with time-dependent thresholds). Its value is V_t^{KS} ; by construction $V_0^{\text{KS}} \leq J^{\text{KS}\theta}$ for every θ . (iii) *Kill-switch with verification* KSV^* and *kill-switch with throttle* KST^* : the same with *verify*, respectively *throttle*, added to the restricted set; their values are V_t^{KSV} and V_t^{KST} . (iv) *Paper 2's one-shot rule* P2: at $t = 0$, with belief p_0 , *verify* if p_0 lies in Paper 2's verification region S (Paper 2, Definition 3, with delay cost $r_K\tau$ replaced by $\ell_v r_K$), then take the better terminal action at the posterior; otherwise take the better terminal action at p_0 . No order flow is used. (v) *Observability-blind throttle policy* NB: the optimal policy of Definition 7 computed with $\kappa = 1$, deployed in the model with the true κ .

The best Bayesian kill-switch KS^* is the strongest kill-switch the model allows: it chooses optimally when to stop, and on which side, from the same belief as the graded policy, but may not throttle or verify. The restricted policies KSV^* and KST^* add one rung each, which splits the gain of the full ladder into a verify share and a throttle share. The observability-blind policy NB is the operator who plans as if throttling kept full discriminability.

Definition 10 (Operating characteristics, gains, critical discriminability factor). For a policy π and prior p_0 : the *false-kill rate* $\text{FPR}^\pi = P^\pi(\text{kill} \mid F)$; the *missed-containment rate* $\text{FNR}^\pi = P^\pi(\text{no kill} \mid K)$; the *expected time to a terminal action*; and the *expected throttled time*. The *graded gain* is $\Gamma(p_0) = V_0^{\text{KS}}(p_0, 0) - V_0(p_0, 0) \geq 0$, with the decomposition into the verify share $V_0^{\text{KS}} - V_0^{\text{KSV}}$ and the throttle share $V_0^{\text{KS}} - V_0^{\text{KST}}$. The *value of throttling* at discriminability factor κ is $G(\kappa) = V_0^{-\text{thr}}(p_0, 0) - V_0(p_0, 0; \kappa) \geq 0$, where $V^{-\text{thr}}$ is the optimal value without *throttle* (it does not depend on κ). The *critical discriminability factor* is $\kappa^* = \inf\{\kappa \in [0, 1] : G(\kappa) > 0\}$ ($\kappa^* = 1$ if the set is empty).

Every comparison is in expected cost at the prior, and every policy is reported with its false-kill and missed-containment rates, the expected time to a terminal action and the expected throttled time beside its cost. *Dominance* of a policy over another means strictly lower expected cost at the prior, $\Gamma(p_0) > 0$ for the graded gain; it is claimed only on a stated region of parameters, never for all of them.

3.6 Assumptions

Assumption 1 (Cost orderings). $H_K > C_K$ and $C_F > H_F$ (Paper 2, Assumption 1); in the three-state model also $C_B > H_B$. Stage costs satisfy $r_K \geq 0$, $\delta \geq 0$, $c_v \geq 0$, $\rho \in [0, 1]$.

Assumption 2 (Conditionally i.i.d. Gaussian order flow). Given M and the postures taken, the per-period order-flow statistics are independent, with the laws of Definition 3; the separation d does not change with time. (A time-varying separation d_t is a stated variant, not the baseline.)

Assumption 3 (Separate channel). The verification outcome Z is conditionally independent of all order-flow statistics given M (Paper 2, Assumption 2), and $s + t > 1$.

Assumption 4 (Throttle effects). $0 \leq \kappa \leq 1$: throttling never makes the causes easier to tell apart. This is a restriction of Paper 3’s model, not a consequence of Paper 1’s: Proposition 4(iv) gives instances with $\kappa_W > 1$. ρ and κ are treated as independent parameters; the linear-model relations between them (Proposition 4) are stated where used.

Assumption 5 (Known model; non-strategic cause). All parameters (d , κ , ρ , δ , s , t , c_v , ℓ_v , the loss table, the prior) are known to the operator, and the cause does not react to the operator’s actions (a strategic attacker is Paper 4’s scope).

Assumption 2 treats the per-period statistics as independent with a constant separation. In the linear-Gaussian model of Paper 1 the separation can grow with time through state feedback; a time-varying separation d_t is a stated variant. Assumption 4 excludes throttles that make the causes easier to tell apart; Section 3.7 shows that it is a restriction of the model, not a consequence of Paper 1’s.

Remark 1 (Equivalent-Gaussian calibration). Paper 1’s ABIDES constructions were never shown to be Gaussian. A separation backed out of a Paper 1 accuracy through the Gaussian benchmark $\text{Acc}^* = \Phi(d_W/2)$ is an *equivalent-Gaussian calibration heuristic*, and it is a separation d_W for the classifier’s whole observation window, not a per-period separation. Under Assumption 2, n_W periods of monitoring give separation $d\sqrt{n_W}$, so the heuristic per-period value is $d = d_W/\sqrt{n_W}$, where n_W is the number of decision periods the window is taken to span (a modelling choice, stated wherever used). Every result evaluated at a calibrated d is also reported over a direct sweep of d .

3.7 Where $\kappa < 1$ comes from

The discriminability factor is a parameter of the POMDP. To see what it stands for, we return to the linear-Gaussian model of Paper 1 [3], in which an incident’s injection enters the observation of the book and, under state feedback, its latent state, and model two throttles in it.

Definition 11 (Throttles in Paper 1’s linear-Gaussian model).

Take Paper 1's output-level model (Definition D1) or state-level model (Definition D2) for a pair of mechanism instances (K, θ_K, b_K) , (F, θ_F, b_F) (Definition D3), under Paper 1's assumptions (A1)–(A4) and (A6), with onset τ and $\Delta_0 = \theta_K b_K - \theta_F b_F$. A *throttle schedule* is a set Θ of periods, the throttled periods; $\Theta = \emptyset$ is monitoring throughout. (i) A *scaling throttle* with factor $r \in [0, 1]$ replaces the injection $\theta_M b_M$ by $r \theta_M b_M$ in every throttled period, wherever the injection enters (the observation and, under D2, the state recursion). (ii) A *feedback-suppressing throttle* keeps the injection in the observation and removes it from the state recursion in throttled periods: under D2, $x_{t+1} = Ax_t + w_t$ for $t \in \Theta$ (the throttled periods follow D1). For a finite window W (Paper 1, Definition D4) let $d_W(\Theta)$ be the whitened mean separation of $Y_W = (y_t)_{t \in W}$ under schedule Θ (Paper 1, Definition D11). When $d_W(\emptyset) > 0$, the *window discriminability factor* is $\kappa_W(\Theta) = d_W(\Theta)/d_W(\emptyset)$. Under Assumption 2, a window of n periods in one posture has separation $d\sqrt{n}$ under monitoring and $\kappa d\sqrt{n}$ under throttling, so $\kappa_W = \kappa$: κ_W is the quantity of Paper 1's model that Paper 3's κ stands for.

Proposition 4 (Where $\kappa < 1$ comes from). In the setting of Definition 11, let W be a finite window and $t_W = \max W$. (i) (*Scaling throttle.*) If $\Theta \supseteq \{\tau, \dots, t_W\}$, the mean difference of Y_W between K and F under Θ is r times the one under monitoring, in D1 and in D2; hence $\kappa_W(\Theta) = r$ whenever $d_W(\emptyset) > 0$. Under D1 it suffices that $\Theta \supseteq W \cap \{t \geq \tau\}$. (ii) (*Feedback-suppressing throttle on an output-confounded pair.*) Suppose the model is D2 and $C\Delta_0 = 0$. If $\Theta \supseteq \{\tau, \dots, t_W\}$, then Y_W has the same law under K and F , so $d_W(\Theta) = 0$. Under monitoring, $d_W(\emptyset) > 0$ if and only if $C \sum_{j=1}^k A^j \Delta_0 \neq 0$ for some $k \geq 1$ with $\tau + k \in W$; if $W \cap \{t \geq \tau\} = \{\tau, \dots, \tau + H - 1\}$, this holds if and only if $CA^j \Delta_0 \neq 0$ for some $1 \leq j \leq H - 1$. When it holds, $\kappa_W(\Theta) = 0$. (iii) (*Late start.*) If the throttle starts after the onset, $\Theta = \{t_0, t_0 + 1, \dots\}$ with $t_0 > \tau$, (i) and (ii) can fail. Let $D_{t_0} = \sum_{j=1}^{t_0-\tau} A^j \Delta_0$. Under D2, for $t \geq t_0$, the mean difference of y_t is r times the one under monitoring plus $(1 - r)CA^{t-t_0}D_{t_0}$ for the scaling throttle, and $CA^{t-t_0}D_{t_0} + C\Delta_0$ for the feedback-suppressing throttle. The residual term $CA^{t-t_0}D_{t_0}$, the memory of the unthrottled feedback before t_0 , is non-zero at $t = t_0$ for some (A, C, Δ_0) with $C\Delta_0 = 0$; for those, the feedback-suppressing throttle has $d_W(\Theta) > 0$ for every window $W \subseteq \Theta$ that contains t_0 , and the scaling throttle with $r < 1$ has $\kappa_W(\Theta) \neq r$ for $W = \{t_0\}$. (iv) (*Other pairs.*) If $C\Delta_0 \neq 0$, the factor κ_W of a feedback-suppressing throttle with $\Theta \supseteq \{\tau, \dots, t_W\}$ is not signed by the model: there are instances with $\kappa_W < 1$, $\kappa_W = 1$ and $\kappa_W > 1$. In particular Assumption 4 ($\kappa \leq 1$) is a restriction of Paper 3's model, not a consequence of Paper 1's.

Proposition 4 is proved directly from Paper 1’s model definitions; it uses neither of Paper 1’s theorems. Part (i) ties the two effects of a scaling throttle together: in Paper 1’s model a throttle that scales the injection by r has $\kappa = r$, so the harm factor and the discriminability factor move together. Part (ii) is the sharpest case. For a pair confounded at the output level, $C\Delta_0 = 0$, all the discriminability of monitoring comes from the feedback of the injection into the state, and a throttle that removes that feedback from the onset leaves none: $\kappa = 0$. The condition under which monitoring separates such a pair over a finite window is the negation of the condition of Paper 1’s finite-horizon Theorem 2; this is the sense in which Paper 1’s Theorem 2 is the mechanism of part (ii). Part (iii) qualifies part (ii): an operator throttles after the alarm, not at the onset, and the feedback accumulated before the throttle starts leaves a residual term that decays according to the dynamics. The value $\kappa = 0$ is exact only from the onset, and otherwise holds after the transient. Part (iv) shows that for pairs that are not confounded at the output level the sign of $\kappa - 1$ is not determined by the model, which is why the POMDP sweeps κ over $[0, 1]$ and states $\kappa \leq 1$ as Assumption 4. In the POMDP, ρ and κ are treated as independent parameters; the diagonal $\kappa = r$ of part (i) is a distinguished case.

4 Structure of the optimal policy

This section states what can be proved about the optimal policy for every parameter value. The proofs are given in the supplementary material; they have been checked by the author and have not yet been independently verified. Section 5 then asks what cannot be proved in general: whether the gains these results allow are strictly positive and material on the calibrated region.

4.1 Concavity

Every result of this section rests on one operator: the expected value of a function of the Bayes posterior after one observation.

Lemma 1 (Bayes operator: concavity, continuity, garbling). Let ν be a σ -finite measure on an observation space O and $\lambda_K, \lambda_F \geq 0$ be ν -integrable. For a bounded Borel $f : [0, 1] \rightarrow \mathbb{R}$ and $p \in [0, 1]$ put

$$(T_\lambda f)(p) = \int_O \Lambda_p(o) f(q_p(o)) \nu(do), \quad \Lambda_p = p \lambda_K + (1-p) \lambda_F, \quad q_p = \frac{p \lambda_K}{\Lambda_p},$$

with the integrand read as zero where $\Lambda_p(o) = 0$. When λ_K, λ_F are probability densities, $(T_\lambda f)(p) = \mathbb{E}[f(p')]$ for the Bayes posterior p' after one observation drawn from the predictive law $\Lambda_p \nu$. (i) If f is concave, $T_\lambda f$ is concave on $[0, 1]$. (ii) If f is continuous, $T_\lambda f$ is continuous on $[0, 1]$. (iii) (*Garbling.*) Let λ'_K, λ'_F be probability densities with respect to a σ -finite ν' on O' , and let $\Gamma(o \mid o') \geq 0$ be jointly measurable with $\int_O \Gamma(o \mid o') \nu'(do) = 1$ for every o' . If $\lambda_M(o) = \int_{O'} \Gamma(o \mid o') \lambda'_M(o') \nu'(do')$ for $M \in \{K, F\}$ (the same kernel for both causes), then $T_\lambda f \geq T_{\lambda'} f$ on $[0, 1]$ for every concave continuous f . (iv) (*Gaussian garbling.*) For $0 \leq r < 1$ and $k \geq 1$, the kernel $\Gamma(y \mid y') = \phi_k((y - ry')/\sigma_r)/\sigma_r^k$, $\sigma_r = \sqrt{1 - r^2}$, maps $\mathcal{N}(m, I_k)$ to $\mathcal{N}(rm, I_k)$ for every $m \in \mathbb{R}^k$. (i)–(iii) hold in the three-state model with p replaced by $\pi \in \Delta_2$, $\Lambda_\pi = \sum_M \pi(M) \lambda_M$ and the posterior $\pi_o(M) = \pi(M) \lambda_M(o) / \Lambda_\pi(o)$.

Part (iii) is Blackwell's ordering in the form used here: an observation obtained from another by a noise kernel common to both causes is worth less to an operator whose continuation value is concave. Part (iv) shows that throttling is such a garbling: the Gaussian law of Definition 3 at separation κd is a garbling of the law at separation d .

Proposition 1 (Concavity). Under Definitions 1–7 and Assumptions 2–3, for every $t \in \{0, \dots, T\}$ and $v \in \{0, 1\}$: (i) in the two-state model $V_t(\cdot, v)$ and every $Q_t(\cdot, v, a)$ are concave and continuous on $[0, 1]$; (ii) in the three-state model $V_t(\cdot, v)$ and every $Q_t(\cdot, v, a)$ are concave and continuous on Δ_2 ; (iii) (*grid, two-state model*) if in Definition 8 every expectation $\mathbb{E}[g(p')]$ is replaced by a *Bayes-consistent* rule $\sum_j \omega_j \Lambda_p(o_j) g(q_p(o_j))$ with nodes o_j and weights $\omega_j \geq 0$ that do not depend on p (the operator T_λ of Lemma 1 for the discrete measure $\sum_j \omega_j \delta_{o_j}$), then for every t, v the grid values $(\tilde{V}_t(i/N, v))_{i=0}^N$ form a concave sequence and their piecewise-linear interpolant is concave on $[0, 1]$. No grid concavity is claimed for the per-component Gauss–Hermite rule of Definition 8, which is not Bayes-consistent, nor for the triangular grid of the three-state model, on which piecewise-linear interpolation of concave data need not be concave.

Part (iii) is used in the numerical study as a test of the solver: with the Bayes-consistent quadrature rule, grid values that fail to be concave in the belief reveal an implementation error, not a property of the model.

4.2 Monotonicity in discriminability and the critical factor

Theorem 1 (Monotonicity in discriminability; a single critical factor). Under Definitions 1–7, 10 and Assumptions 2–4, with all other parameters fixed: (i) for $0 \leq \kappa < \kappa' \leq 1$, $V_t(\cdot, v; \kappa) \geq V_t(\cdot, v; \kappa')$ for every t and v , in the two-state and in the three-state model; (ii) in the two-state model,

for $0 \leq d < d'$, $V_t(\cdot, v; d) \geq V_t(\cdot, v; d')$ for every t and v (the monitoring separation is d and the throttling separation κd in both); (iii) at every prior p_0 , the value of throttling $G(\kappa)$ is non-negative and non-decreasing on $[0, 1]$, so $\{\kappa \in [0, 1] : G(\kappa) > 0\}$ is empty, $(\kappa^*, 1]$ or $[\kappa^*, 1]$; (iv) for $\kappa < \kappa^*$, $G(\kappa) = 0$: an optimal policy of the recursion without **throttle** is optimal from p_0 , so an optimal operator need never throttle; for $\kappa > \kappa^*$, $G(\kappa) > 0$ and every policy that is optimal from p_0 throttles with positive probability. (iii) and (iv) hold in the three-state model with p_0 replaced by π_0 .

Theorem 1(iii)–(iv) is what makes the critical discriminability factor κ^* (Definition 10) a single threshold: the value of throttling $G(\kappa)$ cannot switch on, off and on again as κ grows. Below κ^* the throttle has no value, and a policy that never throttles is optimal; above it, every optimal policy throttles with positive probability. The statement is weak in one direction on purpose: the tie-broken policy of Definition 7 may still throttle below κ^* at a belief where throttling ties with killing, at no change in cost, so an optimal operator *need* never throttle there. Theorem 1 does not locate κ^* . Whether κ^* lies strictly inside $(0, 1)$ on the calibrated region, rather than at 0 (throttling is worth it however much it blinds) or at 1 (throttling is never worth it), is claim 2a, which the numerical study tests.

4.3 Kill and leave regions

Proposition 2 (Kill and leave regions). Under Definitions 1–7 and Assumptions 1–3, in the two-state model let $q^* = b/(a + b)$ and, for $t \leq T$ and $v \in \{0, 1\}$, let $\mathcal{K}_t(v) = \{p : Q_t(p, v, \text{kill}) = V_t(p, v)\}$ and $\mathcal{L}_t(v) = \{p : Q_t(p, v, \text{leave}) = V_t(p, v)\}$ be the sets where kill, respectively leave, is optimal. Then: (i) $V_t(1, v) = C_K$ and $V_t(0, v) = H_F$; (ii) $\mathcal{K}_t(v) = [p_t^K(v), 1]$ and $\mathcal{L}_t(v) = [0, p_t^L(v)]$ are closed intervals with $p_t^L(v) \leq q^* \leq p_t^K(v)$, and $\mathcal{K}_t(v) \cap \mathcal{L}_t(v) \subseteq \{q^*\}$; (iii) under the tie-breaking of Definition 7 the optimal policy takes **leave** exactly on $\mathcal{L}_t(v)$, and takes **kill** on the set where kill is the unique optimal action, an interval that is open relative to $[0, 1]$ and contained in $\mathcal{K}_t(v)$; if $r_K > 0$ and $\delta + pr_K > 0$, that interval contains 1. (iv) In the three-state model (with $C_B > H_B$, Assumption 1), the set where kill is optimal is convex and contains the vertex e_K , and the set where leave is optimal is convex and contains the edge $[e_F, e_B]$. Whether **monitor**, **throttle** and **verify** occupy ordered bands between $p_t^L(v)$ and $p_t^K(v)$ is not claimed.

The operator kills at high beliefs and leaves at low ones, and the two regions are separated by the passive threshold q^* . Between them lie the beliefs at which it pays to keep watching, to throttle or to verify. Proposition 2 does not say how **monitor**, **throttle** and **verify** are arranged in that middle

region; the optimal-action maps of the numerical study show it.

4.4 Embedding of the one-shot rule

Proposition 3 (Embedding of Paper 2). Under Definitions 1–7, 9 and Assumptions 1–3, in the two-state model let R and R_v be Paper 2’s passive and verify-first risks (Paper 2, Definitions 1 and 3) with Paper 2’s delay harm $r_K\tau$ read as $\ell_v r_K$, and $S = \{p \in [0, 1] : R_v(p) < R(p)\}$ Paper 2’s verification region (Definition 3 there). Suppose $T \geq \ell_v$. (i) For every $d \geq 0$, the one-shot rule P2 of Definition 9(iv) has expected cost $J^{P2}(p) = \min\{R(p), R_v(p)\}$. (ii) (*Order flow uninformative.*) If $d = 0$, then, whatever ω , $V_t(p, 1) = R(p)$ for every $t \leq T$, and $V_t(p, 0) = \min\{R(p), R_v(p)\}$ if $t + \ell_v \leq T$ and $V_t(p, 0) = R(p)$ otherwise; in particular $V_0(p, 0) = J^{P2}(p)$. (iii) If $d = 0$, $r_K > 0$ and $\delta + \rho r_K > 0$, the optimal policy of Definition 7 (with its tie-breaking) from p is P2: at $t = 0$ it takes verify if $p \in S$, leave if $p \notin S$ and $p \leq q^*$, kill if $p \notin S$ and $p > q^*$, except at beliefs $p > q^*$ with $R_v(p) = R(p)$, where verify and kill are both optimal and Definition 7 breaks the tie towards verify while Paper 2 breaks it towards containing; after verify, at $t = \ell_v$ it takes the better terminal action at the posterior (leave on a tie). If $r_K = 0$ or $\delta + \rho r_K = 0$, (ii) still holds, but the tie-broken policy may monitor or throttle before acting, at the same expected cost. (iv) (*Sequential weakly beats one-shot.*) For every $d \geq 0$, $V_0(p, 0; d) \leq J^{P2}(p)$. If $T < \ell_v$, verify is never available (Definition 7); in particular, at $T = 1$ the embedding requires $\ell_v = 1$.

When order flow carries no information, the sequential problem collapses to Paper 2’s one-shot choice between leaving, verifying and containing, and the optimal value equals the cost of Paper 2’s rule. For every d , the sequential optimum is at least as good as the one-shot rule, by part (iv). Whether it is strictly better when order flow is informative and the horizon allows more than one decision is claim 1b, which the numerical study tests.

4.5 What the results do not give

Weak dominance of the graded policy over the best kill-switch, $\Gamma(p_0) \geq 0$, holds by construction, because KS^* is in the graded policy’s class; it is not a finding. The results above do not say when the gain is strict, where κ^* lies, or how much an observability-blind operator loses. They also say nothing about the arrangement of the middle region beyond Proposition 2. These are the questions of Section 5.

In the three-state model, Lemma 1, Proposition 1(ii), Theorem 1(i),

(iii)–(iv) and Proposition 2(iv) hold on the 2-simplex. Verification reads authorization, which is valid under both a fault and a benign shock: under Definition 4 with $t_B = t$, its outcome has the same law under F and B . Only order flow separates these two causes, so where a compromise has been ruled out the throttle’s observability price is paid in the F – B separation alone. What this changes in the optimal policy is part of claim 4, tested numerically.

5 Numerical study

The results of Section 4 hold for every parameter value but leave open the paper’s substantive claims. This section describes a pre-registered exact computation that tests them. The design, the tolerance, the materiality margin, the calibrated region and the criteria that would support, weaken or refute each claim were fixed before any result existed.

5.1 Questions

On a calibrated region of parameters, computed exactly by backward induction on the belief grid (Definitions 1–10):

1. Is the graded gain $\Gamma(p_0)$ strictly positive and material, and how much of it is the throttle’s share rather than verification’s? (Claim 1.)
2. When order flow is informative, is the gap between Paper 2’s one-shot rule P2 and the optimal graded policy material? (Claim 1b.)
3. Where is the critical discriminability factor κ^* : strictly inside $(0, 1)$, at 0 or at 1? (Claim 2a.)
4. For which κ does the observability-blind policy NB cost more than the best kill-switch KS*? (Claim 2b.)
5. With a benign third cause, does modelling it change the optimal policy and its cost materially? (Claim 4.)

5.2 Calibrated region

The region is built from Paper 1’s cost study [3] and a pre-registered box for what Paper 1 does not determine.

- *Cells.* The cells in which Paper 1’s classifier does not beat the best constant decision. Each cell supplies its loss table (C_K, C_F, H_K, H_F) , with the measured cost of leaving a fault, and its prior p_0 . The remaining cells of Paper 1’s table are computed as a contrast set, reported beside the calibrated region and not part of it.
- *Separation.* The equivalent-Gaussian calibration (Remark 1) fixes the per-period separation under monitoring, $d = d_W/\sqrt{n_W}$; throttling multiplies it by κ , which is swept over $[0, 1]$. The window separation d_W is backed out of the cell’s classification accuracy, and the number of periods n_W the window spans is a modelling choice, taken at several values. Because this calibration is a heuristic, d is also swept directly, beside it and outside the calibrated region.
- *Pre-registered box.* The horizon T ; the harm rate r_K , set as a multiple of a/T so that total harm over the horizon is measured against what killing saves; the harm factor ρ ; the disruption cost δ , set as a multiple of b/T ; the verification sensitivity and specificity $s = t$, its cost c_v , set as a fraction of $\min(a, b)$, and its delay ℓ_v , with no order-flow observation during the delay ($\omega = 0$, the analogue of Paper 2’s no-further-evidence assumption).
- *Sensitivity, outside the calibrated region.* Other horizons, a longer verification delay, $\omega = 1$, asymmetric $s \neq t$, other verification costs, and a zero cost of leaving a fault.
- *Three-state set.* The calibrated cells with $C_B = C_F$ and $H_B = H_F$, and a scalar order-flow statistic in which the benign shock’s mean is placed relative to the fault’s so that the F – B separation is a swept fraction of d ; the share of the non-compromise prior mass on B is also swept.

5.3 Metrics and decision criteria

Each point is measured against two reference costs: the *oracle cost* $O(p_0) = p_0 C_K + (1 - p_0) H_F$ of an operator who knows the cause, and the *residual ambiguity* $\Delta(p_0) = V_0^{\text{KS}}(p_0, 0) - O(p_0)$, what the best kill-switch still loses to perfect information.

- *Grid convergence.* A point is grid-converged (Definition 8) if doubling N and n_q changes every reported value by at most a tolerance proportional to $\Delta(p_0)$. A difference of two converged values within twice that tolerance is zero up to the grid tolerance.

- *Materiality.* The graded gain is material at a point if it recovers at least a fixed share of the residual ambiguity; the gap to P2 is material if it recovers the same share of what P2 loses to the oracle.
- *Shares.* The throttle share $V_0^{\text{KS}} - V_0^{\text{KST}}$, the verify share $V_0^{\text{KS}} - V_0^{\text{KSV}}$ and their interaction are reported beside Γ .
- *Critical factor.* At each point, κ^* is classed as 0, 1 or interior from G at the ends of the sweep, and located by bisection when interior. A decrease of G in κ beyond the tolerance would contradict Theorem 1 and is treated as an error of the solver, not as evidence.
- *Blind operator.* κ_{NB} is the largest κ of the sweep at which J^{NB} exceeds V_0^{KS} beyond the tolerance.
- *Beside cost.* False-kill and missed-containment rates, expected time to a terminal action and expected throttled time, for every policy; optimal-action maps over belief and time for a representative point of each cell, and on the simplex for the three-state set.

Each claim is supported, weakened or refuted by criteria fixed in advance. Claim 1 is refuted if the graded gain is zero up to the tolerance at every point, and weakened if it is material at no more than half of the points, if it comes entirely from verification, or if its material points lie only outside the calibrated region. Claim 2a is refuted if $\kappa^* = 0$ at every point or $\kappa^* = 1$ at every point, and weakened if κ^* is interior only outside the calibrated region. Claim 2b is refuted if NB never costs more than KS* beyond the tolerance, and weakened if it does so only outside the calibrated region. A claim is *inconclusive* if more than a quarter of its points are not grid-converged after two doublings or straddle the materiality threshold. Negative outcomes are reported beside positive ones.

5.4 Checks before any result

Three results of Section 4 are checked on the solver’s output before the sweep is read: concavity of the two-state grid values under the Bayes-consistent rule (Proposition 1(iii)); monotonicity of G in κ (Theorem 1); and, at $d = 0$ and $T = \ell_v$, equality of the optimal value with the cost of Paper 2’s rule, with S computed on the grid from Paper 2’s definition directly (Proposition 3). In the three-state model, the value of verification on the F – B edge must be zero up to the tolerance. A failure of any of these is an error of the implementation.

5.5 Results

The results of the sweep will be reported here from the generated tables of brief B001, beside the design above, with negative points beside positive ones.

6 Discussion

What is claimed, and at what strength. The structural results of Section 4 hold for every parameter value. The paper’s two substantive claims do not: strict dominance of the graded policy over the best kill-switch, and an interior critical discriminability factor, are claims about a calibrated region, and they stand or fall with the pre-registered computation of Section 5. Weak dominance is not a finding, since the kill-switch is one of the graded policies. The observability price is stated in two forms that do not collapse into each other. For an optimal operator it is a threshold: below κ^* the throttle has no value (Theorem 1), which says when to throttle, not that throttling is harmful. For an operator who plans as if throttling kept full discriminability, it is a loss: the policy NB is tested against the best kill-switch, and a region where it does worse would make ignoring the price costly, not merely suboptimal.

The discriminability factor. The POMDP treats ρ and κ as independent parameters and κ as a constant. Proposition 4 shows that neither simplification is innocent in Paper 1’s linear model: a scaling throttle ties κ to the scaling factor, and a feedback-suppressing throttle started after the onset has a transient in which the residual feedback keeps the throttled observations informative. The constant κ of the POMDP is exact for a throttle in force from the onset and is otherwise an idealisation that holds once the transient has decayed. Throttles that make the causes easier to tell apart, for instance because a persistent adversary pushes against the limit, are excluded by Assumption 4; they would turn *throttle* into an information-gathering action.

Calibration. The separation is backed out of a classifier accuracy through a Gaussian benchmark that Paper 1’s simulated constructions were never shown to satisfy, and it is a separation for a whole window, not for a period. It is therefore labelled a heuristic, converted to a per-period value under a stated window length, and accompanied by a direct sweep of d . Assump-

tion 2 also ignores the growth of separation with time that state feedback produces in Paper 1’s model.

Known model, non-strategic cause. The operator knows every parameter and the cause does not react (Assumption 5). The contribution is the structure of the trade-off between harm, disruption and evidence, which needs exact solutions; robustness to a misjudged κ is what the blind policy NB measures. Learning the model online, and an adversary who adapts to the operator’s actions, are outside the paper’s scope; the strategic case is the subject of Paper 4.

More than two candidate causes. The three-state extension is the one place where verification is structurally partial: it separates a compromise from the other two causes but not a fault from a benign shock. It is solved exactly on the simplex, with the structural results that carry over (Section 4.5). Whether it changes the policy materially, or only repeats the two-state picture on a larger grid, is claim 4 of the numerical study; if it does not, the extension is reported as a method, not a result.

7 Conclusion

Responding to an ambiguous incident on a trading gateway is a sequential decision in which containment has a price in evidence. We have modelled it as a POMDP over the hidden cause, with a graded ladder of actions and a throttle that reduces harm and discriminability together, and solved it exactly on a belief grid. The value function is concave, the kill and leave regions are intervals, and the optimal value is monotone in the discriminability factor, so that there is a single critical factor κ^* below which an optimal operator need never throttle. Paper 2’s one-shot verification rule is the special case with uninformative order flow. In Paper 1’s linear-Gaussian model, a throttle that suppresses the incident’s feedback from the onset leaves no discriminability for pairs confounded at the output level, and a scaling throttle keeps exactly its scaling factor. Whether the graded policy strictly beats the best kill-switch on a calibrated region, where κ^* lies there, and what an operator loses by ignoring the observability price are tested by the pre-registered computation of Section 5, with criteria fixed in advance.

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